INCH-POUND

MIL-PRF-26542/7B
w/AMENDMENT 1
14 February 2006
SUPERSEDING
MIL-PRF-26542/7B
16 May 2005

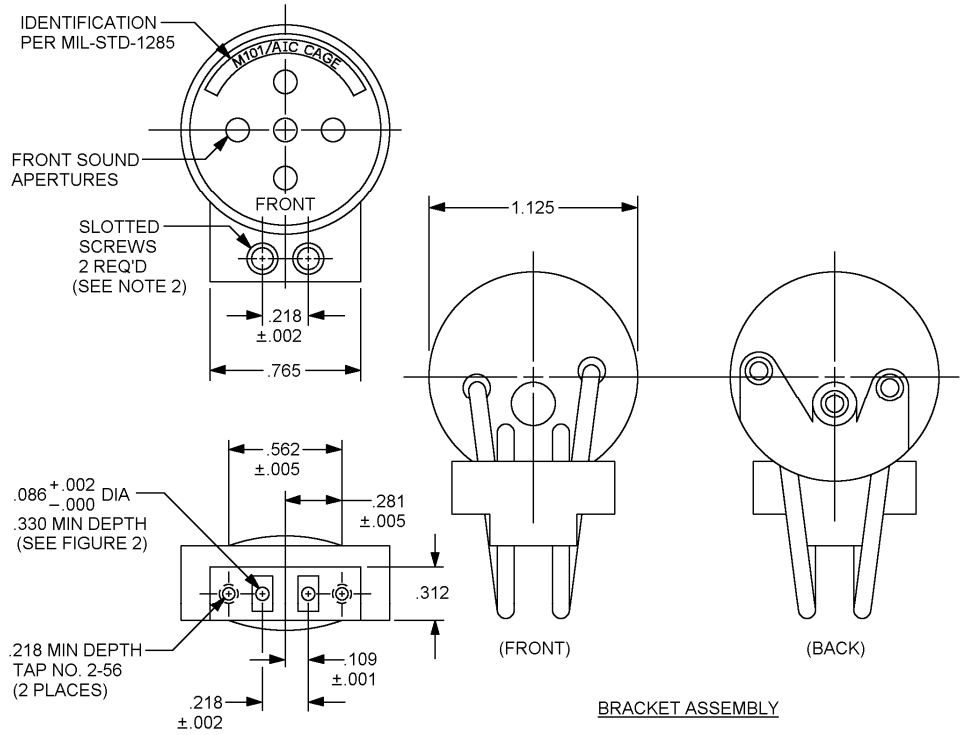
PERFORMANCE SPECIFICATION SHEET

MICROPHONE ASSEMBLY, M-94B/A

Inactive for new design after March 12, 2002.

This specification sheet is approved for use by all
Departments and Agencies of the Department of Defense.

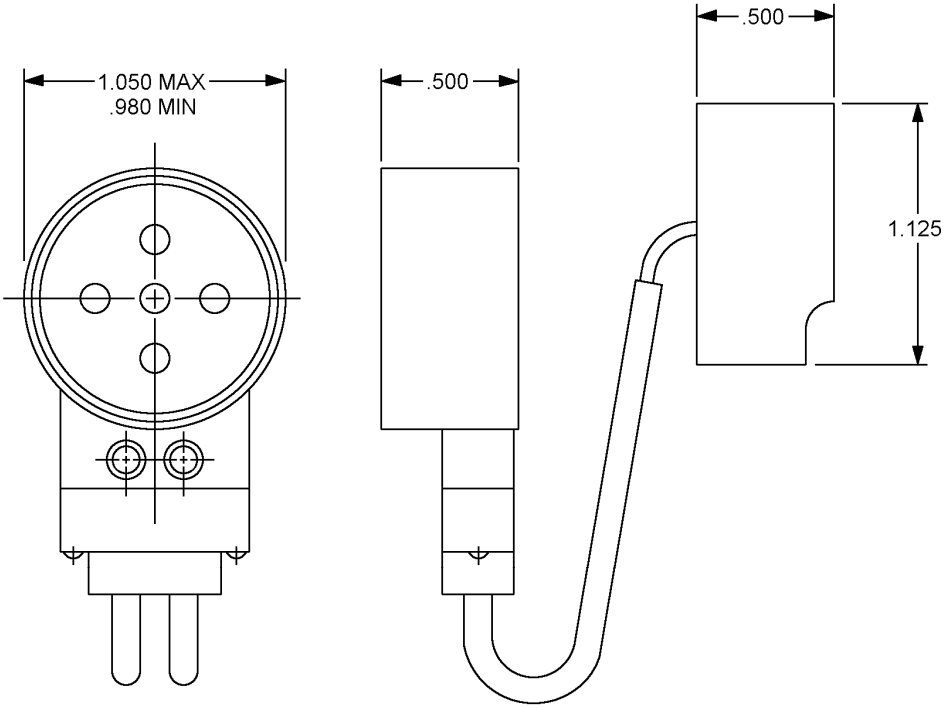
The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-26542.



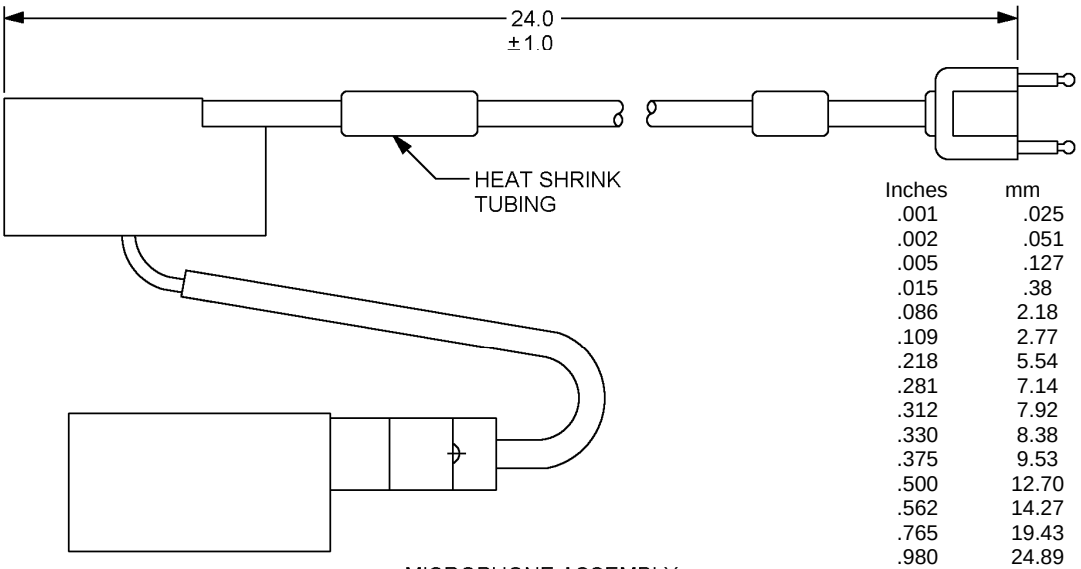
NOTES:
1. Dimensions are in inches. Unless otherwise specified, tolerances are ± .015 (0.38 mm).
2. Set-screws shall hold the part securely to the bracket, shall be slotted (for interchangeability of spares among manufacturers and tri-service applications), and shall be recessed. Screws shall be in accordance with AN500 (part number AN500D2-6, stainless steel, 3/8 (.375) inch length, 2-56 thread), for interchangeability of spares.
3. Hole depths shall be as specified, in order to provide a secure bracket connection which does not compromise the mechanical stability of the microphone base.
4. Sound aperture location and configuration are optional, provided that frequency response, environmental, and performance requirements are met as specified.
5. Metric equivalents are given for general information only.

FIGURE 1. Microphone assembly.

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TRANSMITTER AND BRACKET ASSEMBLY

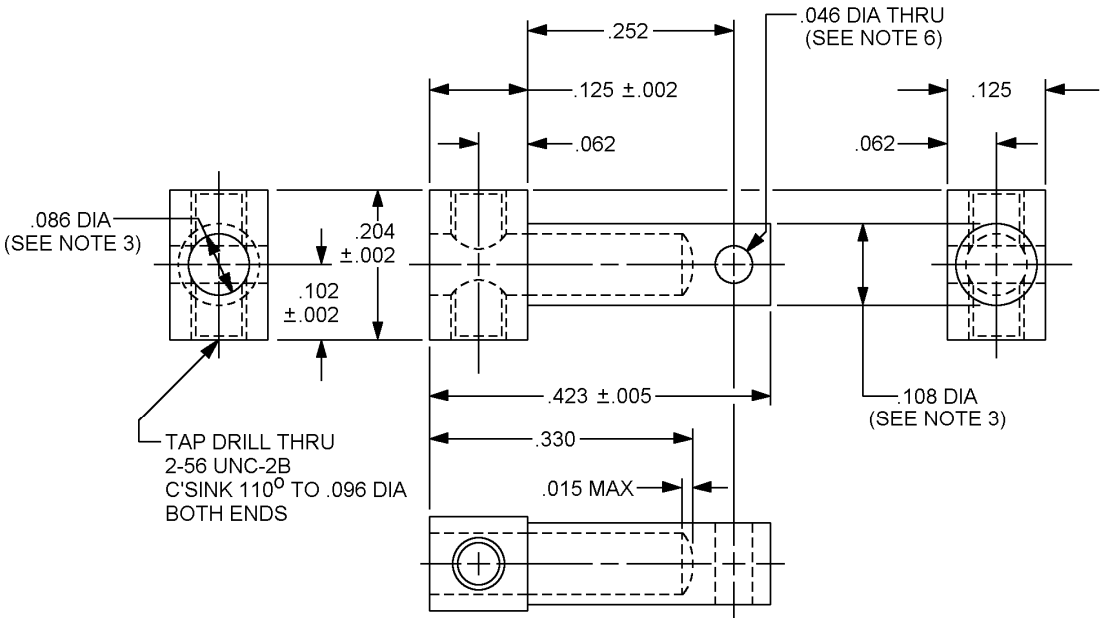


MICROPHONE ASSEMBLY
(SIDE VIEW)

Inches	mm
.001	.025
.002	.051
.005	.127
.015	.38
.086	2.18
.109	2.77
.218	5.54
.281	7.14
.312	7.92
.330	8.38
.375	9.53
.500	12.70
.562	14.27
.765	19.43
.980	24.89
1.00	25.40
1.050	26.67
1.125	28.58
24.00	609.60

FIGURE 1. Microphone assembly - Continued.

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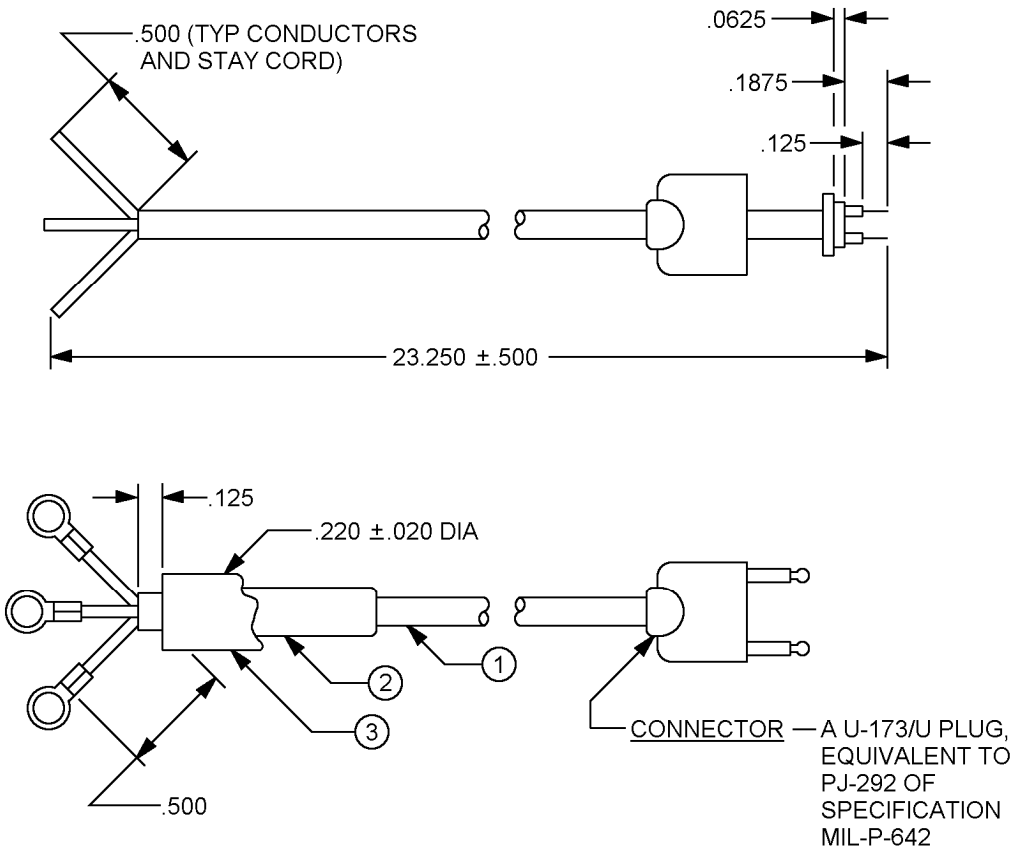


Inches	mm	Inches	mm
.002	0.05	.102	2.59
.003	0.08	.108	2.74
.005	0.13	.125	3.18
.015	0.38	.204	5.18
.046	1.17	.252	6.40
.062	1.57	.330	8.38
.086	2.18	.423	10.74
.096	2.44		

NOTES:
1. Material shall be brass in accordance with ASTM B 16, temper H02 (Half Hard).
2. Silver plate finish in accordance with ASTM B 700, Grade A (MAT, without luster).
3. Diameter to be concentric within .005 inch, Total Indicator Reading (T.I.R.).
4. Dimensions are in inches. Unless otherwise specified, tolerances are ± .003 inch.
5. Metric equivalents are given for general information only.
6. Solder point connection design is at the option of manufacturer.

FIGURE 2. Connector.

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Inches	mm	Inches	mm
.015	0.38	.1875	4.76
.020	0.51	.220	5.59
.0625	1.59	.500	12.70
.125	3.18	23.250	590.55

NOTES:
1. Heat shrink item 2 on item 1, then heat shrink item 3 onto item 2. Caution: Do not heat shrink item 2 and 3 at the same time.
2. Attach stay cord terminal, as close as possible to end of outer jacket.
3. Dimensions are in inches. Unless otherwise specified, tolerances are ± .015 inch.
4. Metric equivalents are given for general information only.

FIGURE 3. Cord assembly.

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REQUIREMENTS:

Color: Semi gloss black, 27038 in accordance with FED-STD-595.

Component parts: See figure 1.

Microphone element: See figures 1 and 2.

Bracket assembly: See figure 1.

Cord assembly: See figures 1 and 3.

Impedance: 4 - 6 ohms.

Resistive load: 5 ohms.

Performance:

Microphone element: One each M101/AIC in accordance with MIL-PRF-26542/4. Screws shall be supplied (see figure 1). The screws shall be supplied with additional hardware necessary to ensure that they do not become disengaged during normal use, such as lock washers.

Weight: The weight of the microphone assembly shall not exceed 1.5 ounces.

Interchangeability: The equipment should meet the interchangeability requirements, as defined in MIL-HDBK-5400.

The microphone assembly shall be tested in accordance with the tests listed in table I.

QUALITY ASSURANCE PROVISIONS:

Extent of qualification:

Qualification testing and approval to MIL-PRF-26542/4 (M101/AIC) shall be sufficient to grant qualification approval to MIL-PRF-26542/7 (M-94B/A).

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TABLE I. Parameter applicability.

Inspection[1/]	Qualification	Group A	Group B	Group C
Group I				
Visual and mechanical inspection	X	X		
Sensitivity at ground level	X	X		
Sensitivity at altitude	X			
Frequency response at ground level	X	X		
Frequency response at altitude				
Impedance	X	X		
Noise cancellation characteristics				
Effect of external magnetic field				
Stray magnetic field	X		X	
Linearity	X		X	
Talk-out				
Dielectric withstanding voltage				
Signal-to-noise	X			
Distortion	X			
Interchangeability				
Group II				
Thermal shock	X			X
Humidity	X			X
Drop	X			X
Pressure equalization	X			X
Explosive decompression				
Salt fog	X			X
Group III				
Vibration	X			X
Bounce				
Altitude	X			X
Moisture barrier seal	X			X
Immersion				

1/ See the associated requirement and method paragraphs within sections 3 and 4 of MIL-PRF-26542, for the above inspections.

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Reference documents. In addition to MIL-PRF-26542, this document references the following:

AN500
ASTM B 16
ASTM B 700
FED-STD-595
MIL-HDBK-5400
MIL-P-642
MIL-PRF-26542/4
MIL-STD-1285

The margins of this specification sheet are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 11
DLA - CC

Preparing activity:
DLA - CC

(Project 5965-2005-002)

Review activities:
Army - AT, AV, CR4
Navy - AS, OS
Air Force – 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

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